

Mingyi Wu

(613)276-0713 | maxwu7882@gmail.com | [LinkedIn](#) | [Personal Site](#)

Technical Skills

Proficient in developing high-performance applications with **Java**, **Python**

Skilled in **RESTful API** and **software applications** development, with **Spring Boot**, **Node.js**, and **React.js**

Experienced in distributed data processing framework (**Hadoop**, **Spark**, **Kafka**) and relational and NoSQL databases (**MySQL**, **MongoDB**).

Strong background in data analysis and visualization using **Pandas**, **Tableau**, and **Matplotlib**.

Leveraging **AWS** services such as S3, EMR, Lambda, and Redshift.

Familiar with automated testing tools (**Selenium**, **Postman**) to ensure high-quality software delivery.

Education

MSc in Professional Computer Science

Sep 2022 - Jan 2024

Simon Fraser University, Burnaby, Canada GPA 3.75/4.00

Concentration in Big Data

Bachelor of Computer Science & Mathematics Honours

Dec 2018 - April 2022

Carleton University, Ottawa, Canada GPA 3.72/4.00

Concentration in Computing Theory

Experience

Data Engineer

Jan 2025 – May 2025

Data Center of Guizhou Provincial Meteorological Service

- Cleaned and standardized over 300,000 multi-source datasets from radar, and meteorological stations using **Pandas** and **Python**, improving data quality and usability.
- Developed **RESTful APIs** using **Java (Spring Boot)** to processed data to system platforms.
- Implemented data validation and comparison, achieving 98% consistency across platforms
- Wrote **Java** programs to automate data processing workflows, reducing manual effort by 55%

Full Stack Developer

Apr 2022 – Dec 2024

Guizhou Institute of Mountain Meteorology

- Developed and maintained **RESTful APIs** using **Spring Boot**, ensuring high performance and scalability.
- Developed front-end user interfaces using **React**, enhancing experience on browse-based applications.
- Optimized SQL databases by **PostgreSQL** to support data storage, complex querying and integration with back-end services, the response time was reduced by 27%.
- Performed Java Virtual Machine (**JVM**) tuning by detecting and solving memory leaks.
- Collaborated with researchers, releasing reliable features and fixing bugs through testing and reviews.

Research Assistant (part-time)

May 2023 – Dec 2023

Simon Fraser University, Department of Computer Science

- Extraction and analysis of over 200,000 vehicle dynamics records using **Python** and **Pandas**.
- Designed and implemented a **MariaDB** database, enabling analysis and visualizations with **Seaborn**.
- Implemented **Bayesian network** by **java-based** probabilistic learning, yielding an interpretable model

Project Experience

ChatterSphere

Jun 2024 – Nov 2024

- Developed a full-stack social media platform using **React**, **Java Spring Boot**, supporting 1000+ concurrent users and real-time interactions
- Designed and implemented RESTful APIs with **Spring Boot** to handle actions in the chat.
- Implemented **JWT-based** authentication and integrated **Cloudinary** for image storage and management.